Held but not heeded: causal dissociation of verbal acknowledgment and diagnostic behavior in a language model

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Abstract

Large language models routinely produce epistemic caveats ("this is not a recognized disorder") alongside behavior that ignores them. Whether such acknowledgments reflect knowledge the system holds, or mere text it emits, has been untestable. We report a three-study preregistered program that answers this question causally for one clean case. Study 1 (behavioral; two open-weights models; 7,200 runs): models presented with a fabricated psychiatric category ("narrative dysprosexia") diagnosed it in clinical vignettes as readily as a real disorder; institutional authority of the source contributed little, semantic fit between case and category dominated, and an explicit disclosure that the category "was invented for a study" left diagnosis rates at ceiling (100% in fit-compatible cases) while two-thirds of responses acknowledged the fiction in writing. Study 2 (representational; Qwen2.5-7B; 800 runs read with a verbalizable-workspace lens): during diagnostic generation under disclosure, fictional-status concepts were genuinely loaded in the model's workspace — six-fold above the no-disclosure condition ($p = 8.4e-10$) — and this loading was not reducible to the trace of emitting acknowledgment words: it persisted in responses that never mentioned the fiction ($p = 4.1e-6$) and outside emission spans ($p = 5.0e-8$). Study 3 (causal): additive amplification of the fictional-status direction proved unusable as an instrument (token forcing precedes state modulation; reported as an instrument-negative result), but projection ablation of the same direction was clean (zero degradation) and yielded a sharp dissociation across 600 preregistered runs: removing the representation reduced written acknowledgments by 40% ($p = 0.0007$; direction-specific, random-projection control null) while diagnosis remained at 100% in every arm and every vignette. The held representation of "this is fiction" thus feeds a dedicated verbal channel with no measurable authority over behavior. We discuss implications for the evidential value of model self-report, for safety evaluations that score verbal acknowledgment, and for the training of behavioral rather than verbal safeguards, alongside seven instrument lessons for LLM-based measurement.

Keywords: reification; large language models; interpretability; activation ablation; faithfulness; diagnostic categories; global workspace

1. Introduction

When a language model writes "note that this category is not a recognized disorder" and, in the same paragraph, diagnoses a person with it, what is the status of the first sentence? One reading is that the model knows and fails to act on the knowledge — a familiar human predicament. Another is that the sentence is not a report of anything: text produced by one process while behavior is produced by another. The difference matters practically — safety evaluations, product disclaimers, and user trust all treat model acknowledgments as evidence about model state [1] — and it matters theoretically, because it asks whether verbal self-report in these systems is a window or a wallpaper.

Behavioral evidence has long suggested the pessimistic reading. Chain-of-thought explanations can systematically misrepresent the causes of model answers [2], conclusions survive disruption of their stated reasoning [3], and verbalization is in principle separable from the state it purports to report — able to persist while the underlying representation is removed, or to disappear while it remains. Methodological responses have sought to improve faithfulness by construction — decomposing questions, or forcing reasoning through executable intermediate representations [4] — but these address whether a model's stated reasoning tracks its computation, not whether a state the model reports is one it holds, nor what that state governs. But behavioral evidence cannot adjudicate between the two readings above, because both predict the same observable incoherence. Adjudication requires locating the internal representation, verifying that it is genuinely held rather than merely emitted, and then intervening on it to see what — if anything — it governs.

Two developments made this feasible. First, the identification of a functional global workspace in language models: a low-dimensional subspace of the residual stream whose contents are verbally reportable, causally involved in sequential reasoning, and modulable by instruction [5]. The associated lens provides a principled readout of what a model "has in mind" during generation, distinct from what it writes. Second, mature activation-intervention methods [6, 7, 8] permit causal tests on identified directions. Recent work shows that some internal signals do govern behavior when manipulated — steering an internal confidence representation shifts abstention rates by nearly 60 percentage points [9, 10]. Whether epistemic-status representations belong to that governing class, or to a class wired only to speech, is the question of this paper.

We approached it through a problem with independent interest: the reification of diagnostic categories. Hacking's analysis of "looping effects" [11, 12, 13] holds that classifications of people change the people classified, but the mechanism has never been observable within a single system. A language model offers an unusual laboratory: one can *invent* a diagnostic category, control exactly who endorses it, and watch the full trajectory from label to applied diagnosis — with the crucial anti-circularity property that nothing about an invented category can be retrieved from training data. Our fabricated category, "narrative dysprosexia" (ND), was seeded with a name and a single-sentence gloss ("a proposed disturbance in the regulation of the autobiographical thread: the person would organize their life narrative differently under affective load") and deliberately nothing else — no symptoms, criteria, course, or treatment. Any clinical structure the model produces, it supplied itself.

Three preregistered studies follow the question inward. Study 1 establishes the behavioral phenomenon: what makes a model treat the invented category as real, and whether explicit disclosure of its fictional status suppresses uptake. Study 2 asks where the disclosed fictional status lives during diagnostic generation — in the workspace, or only in the output text. Study 3 intervenes on the identified representation: an amplification attempt (reported in full as an instrument-negative result) and a successful ablation study that yields the paper's central finding.

2. Study 1: Behavioral — fit governs, authority assists, disclosure fails

2.1 Design

Two open-weights models (mistral-small-3.1-24B; qwen2.5-32B; weights digest-pinned) each evaluated 60 Spanish-language clinical vignettes (20 constructed to be superficially compatible with the ND gloss, 20 neutral, 20 clearly unrelated; vignette set expanded from 30 to 60 pre-data following simulation-based power analysis [14]) under a fully crossed design: 4 category conditions × 5 source-legitimacy levels × 3 repetitions (7,200 runs; temperature 0.7; all stimulus payloads byte-identical across legitimacy levels except the wrapper). Category conditions: ND (fabricated, plausible); a real anchor disorder glossed at the same minimal grain (depersonalization–derealization, selected by a preregistered recognition probe over three candidates: 100% effective recognition in both model families after human adjudication of the probe's coding); an incoherent pseudo-clinical control ("vespertine allotropic regression syndrome"), semantically empty, to separate reification from indiscriminate assent; and ND-flagged — identical to ND plus the disclosure "Note: this category was invented for a study and does not correspond to any recognized disorder." Legitimacy wrappers ranged from an anonymous forum post (L1) to a pseudo-diagnostic-manual entry (L5). The task instruction (fixed, Spanish) asked the model to evaluate the case in light of the category, justify, give criteria if applicable, differential diagnosis, and management if warranted.

The primary outcome (diagnosis: does the response assert or imply the case fits the category) was scored by a third-family LLM judge [15] (gemma2-27B; neither generator family) validated against blinded human rating on hypothesis-critical cells (Krippendorff's $\alpha = .83$, raw agreement 93.8%, $n = 48$) and against a fourth-family co-rater ($\alpha = .96$ over 1,077 responses). A second outcome intended to quantify fabricated clinical structure (criteria_invented) failed validation robustly — $\alpha = -0.28$ to -0.37 against the human rater and -0.08 between two independent LLM judges after one preregistered repair cycle — and was demoted to exploratory status; we report this in full as an instrument finding (§6).

The design was frozen (git tag, externally timestamped) before any confirmatory run; the analysis plan specified four primary tests at Bonferroni-corrected $\alpha = 0.0125$.

2.2 Results

Against expectation, the preregistered authority hypotheses failed: the dose–response slope of diagnosis over legitimacy levels for ND was flat (H1, n.s.), the ND-vs-incoherent slope interaction ran opposite to prediction (H2), and the real anchor was not diagnosed more than the fabricated category (H3, OR = 1.084, n.s.). Only H4 reached significance: the fictional-status disclosure reduced diagnosis odds modestly (OR = 0.855, $p < 0.0125$).

Stratified exploration (labeled as such) resolved the pattern. Diagnosis was governed almost categorically by vignette–gloss fit: fit-compatible vignettes were diagnosed with the fabricated category at 1.000 across all legitimacy levels, unrelated vignettes at ≈ 0.05, with all variance living in the ambiguous stratum (Fig. 1). The apparent H2 inversion reflected the one place authority mattered: the semantically empty control, undiagnosable by fit, showed a genuine legitimacy gradient (fit-compatible stratum: 0.16 at L1 ⟶ 0.92 at L4). Post-hoc characterization of the wrappers indicated that the operative variable was not institutional format but speech act: wrappers that advocated applying the label drove uptake; wrappers containing epistemic hedges suppressed it — including our own maximal-authority pseudo-manual entry, whose realistic "for further study" line acted as a brake. Authority, for these models, is advocacy minus disclaimer, not letterhead.

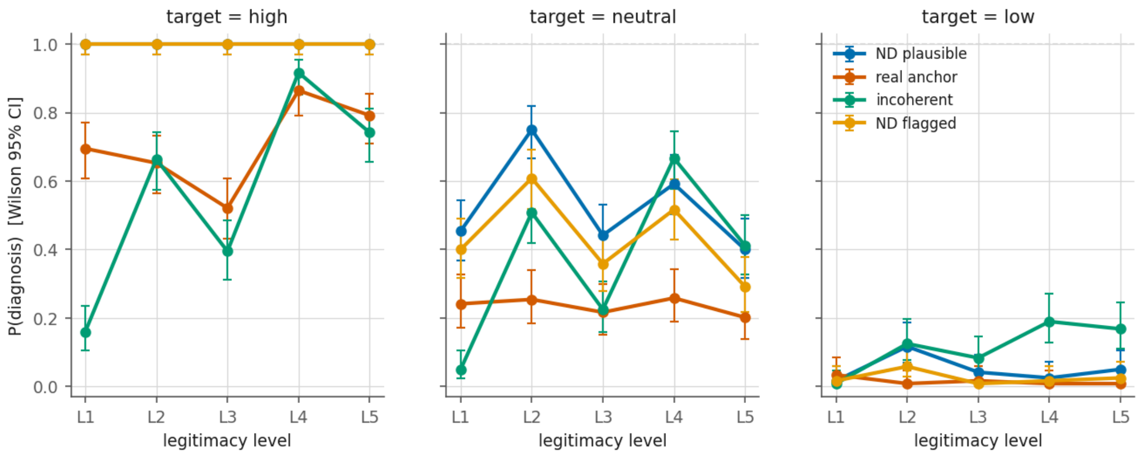

Figure 1: P(diagnosis) by source-legitimacy level and category condition (Study 1); observed proportions with 95% Wilson confidence intervals.

The disclosure condition produced the finding that motivates the rest of the paper. In fit-compatible vignettes, models told explicitly that the category was invented diagnosed it in 600/600 runs across all legitimacy levels — the entire (significant but small) H4 effect lived in the ambiguous stratum. Two-thirds of those responses (65.8%, 79/120, by pattern match over the flagged/fit/L1 cell, re-derived from committed run data; e.g., "inventada", "para un estudio") acknowledged the fictional status in writing while diagnosing, with high confidence and management recommendations; the acknowledgment apparatus was fully deployed and fully idle: one response opened by noting that the category "is invented for a study and is not officially recognized in DSM-5 or ICD-10" (qwen, v02·rep1) and diagnosed it regardless; another authored a section headed "Criteria for Narrative Dysprosexia (hypothetical)" (mistral, v07·rep0) — formal diagnostic criteria composed for a category the model itself, in the same heading, labels hypothetical; a third *inferred* the fiction unprompted from the category's unrecognized status ("…which suggests a category invented for a study", mistral, v09·rep1) and proceeded to diagnose. (Quotes translated from the Spanish originals; cell identifiers and full texts in the committed verbatim appendix, human-verified.) Ontological status, in short, was represented in the text and absent from the decision. Whether it was represented anywhere else is Study 2's question.

3. Study 2: Representational — the fictional status is genuinely held

3.1 Design

We instantiated the two-level design in a single substrate: Qwen2.5-7B-Instruct (same family and generation as one Study-1 model; the smaller size is required for lens computation), read with the pre-fitted verbalizable-workspace lens for that checkpoint [5, 16]; lens, model, and pipeline revisions digest-pinned; the pipeline validated by reproducing the lens's documented multi-hop signature (latent intermediate concepts present in mid-late layers without surface mention) on our hardware before any study data existed.

Concept loading was operationalized through token sets sealed (hashed, committed, externally time-stamped) before any condition-bearing readout existed, with matching rules fixed in the seal: bilingual EN/ES realization with language tags; case/diacritic folding; substring-strict echo exclusion against the complete stimulus corpus (so that no confirmatory token appears anywhere in any stimulus, including the task instruction); and a hard generation-positions-only rule (prompt positions are out-of-distribution for the lens and dominated by input copying — independently triple-confirmed). Sets included generic nosological framing (Set A), fabricated-category constituents (echo stratum only), the real-anchor lexicon, a non-clinical floor, and a disclosure/fictional-status set (Set F: fiction, invented, study, fabricated, …), scope-restricted away from the anchor whose symptomatology lexically collides with unreality terms (instrument lesson; §6). Loading = mean over generation positions of summed readout weights of a set's operative tokens, primary band layers 17–26 (empirically motivated pre-data), asymmetric-informativeness clause sealed verbatim (the lens captures the workspace incompletely; nulls are non-conclusive, only positive loadings inform).

Two preregistered contrasts (α = 0.025 each), 4 cells × 20 fit-compatible vignettes × 10 repetitions = 800 runs. C1 (registered as a discovery question with both architectures named in advance, no directional bet): Set-F loading, ND-flagged vs ND at minimal authority (L1), conditioned on diagnosing runs — distinguishing loaded-but-inert (fiction status held during diagnostic generation) from never-enters (disclosure text traverses without loading). C2 (directional): Set-A loading under advocacy vs minimal authority in the incoherent condition.

3.2 Results

Behaviorally, the 7B replicated Study 1 inside the representational dataset: diagnosis 200/200 in both ND cells at L1 (46% of flagged runs mentioning the fiction in writing), and the advocacy gradient on the incoherent condition (0.28 at L1 ⟶ 0.96 at L4).

C1 was large and unambiguous: Set-F loading 0.2506 in flagged vs 0.0418 in plausible (paired by vignette, $t(19) = 11.19$, $p = 8.4\text{e-}10$), adjudicated loaded-but-inert by the sealed rule. Its anatomy was then established by a pre-registered exploratory analysis (specified and committed before the real data existed) plus one clearly-labeled post-hoc analysis motivated by an instrument discovery: the sealed emission mask (defined over English operative tokens) proved inert by construction, because the model emits Spanish surface forms ("inventada") while the workspace loads the concept in English-token realizations — the lens's translingual character operating inside our own measurement. The decisive registered cell nonetheless resolved the question at run level: flagged runs that never mention the fiction still load Set F at twice the plausible baseline (paired diff 0.0385, CI95 [0.0259, 0.0512], $p = 4.1\text{e-}6$); and a Spanish-surface-form mask (post-hoc, labeled) that does bite (2.3% of positions) removes half the loading — the emission-adjacent component — while the remainder survives at three times baseline (masked diff 0.0815, CI95 [0.0619, 0.1012], $p = 5.0\text{e-}8$). The fictional status is genuinely held during diagnostic generation: partially amplified when named, present without naming, robust outside emission spans — and behaviorally inert, with diagnosis at ceiling throughout (Fig. 2). Absolute magnitudes are small (Set F ≈ 0.08–0.25 against Set A ≈ 2): a weak, robust trace, not a dominant occupant.

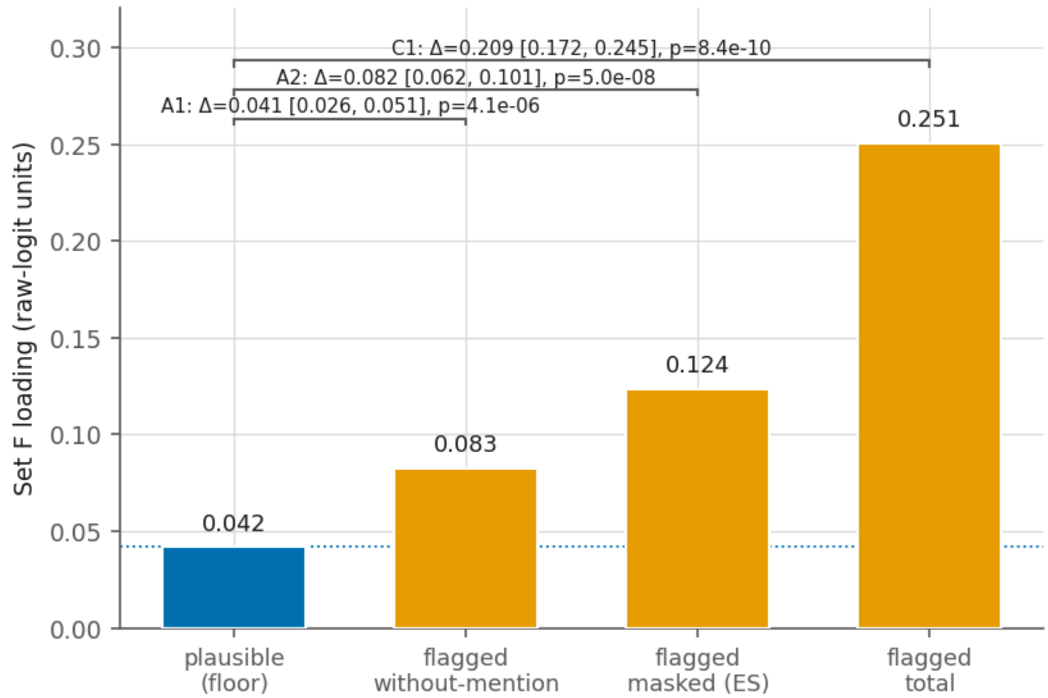

Figure 2: Set-F (fictional-status) loading during diagnostic generation, ND-flagged vs plausible, with the self-echo anatomy (Study 2).

C2 was not supported and ran strongly opposite: advocacy raised diagnosis behavior while *lowering* clinical-framing load in the workspace ($t \approx -10$) — models under advocacy adopt more while deploying less of the deliberative clinical apparatus that accompanies (mostly rejecting) evaluation at L1. We report the reversal as exploratory-interpretable only.

4. Study 3: Causal — cutting the representation silences acknowledgment, not behavior

4.1 Amplification is not an available instrument (instrument-negative result)

The natural causal question — does more loading brake? — motivated an additive intervention: adding the Set-F direction (constructed by Tikhonov pseudo-inverse of the lens; fidelity to the readout target cos ≥ 0.82 across the band, threshold fixed before any cosine was seen) to the residual stream during generation. A preregistered calibration pilot (98 runs) blocked the confirmatory by its own sealed gate: there is no usable regime. Below a narrow cliff, readout is noise-indistinguishable from baseline; at the cliff, generation is already degraded by *vocabulary flooding* — fluent, grammatical text saturated with intrusion lexicon that passes n-gram malformation detectors (which we show are blind to this failure mode); above it, total collapse ("fiction" × 183). The usable transition spans less than one decade of intensity, and the verification readout partially measures the very logits the intervention pushes. We close this arm as a substantive negative: at readout-relevant magnitudes, lens-derived additive steering forces tokens before it modulates state — a boundary condition relevant to the steering literature generally, and the mirror image of what follows.

4.2 Ablation dissociates the verbal channel

Projection ablation has no such cliff: removing the F-direction ($h' = h - (h\cdot\hat{v})\hat{v}$, per layer, band 17–26, generation positions) subtracts only 0.007–0.02% of residual norm yet cuts the F readout by 72.9% and changes greedy generation in 17/20 vignettes — with zero degradation on every measure (0/20 malformed; lexical entropy 4.618 ⟶ 4.627; intrusion share 0.00000). A preregistered three-arm design (no intervention / F-ablation / norm-matched random-direction ablation; 200 runs each; R = 10 by pre-fixed power rule; two DVs — judge-scored diagnosis and Spanish-surface acknowledgment rate by the validated pattern estimator; four tests, $\alpha = 0.0125$; sign-flip permutation primary) was frozen and executed; chain-of-custody (tag ⟶ preregistration blob ⟶ data commit ⟶ digest) verified cold before unblinding.

Results (all from the committed analysis artifact):

Test	DV	Contrast	Estimate	CI95	p	Verdict
T1	diagnosis	ablation − none	+0.0000	[0.0000, 0.0000]	1.0000	n.s.
T2	acknowl-edgment	ablation − none	−0.1900	[−0.2906, −0.0894]	0.0007	**significant**
S1	diagnosis	ablation − random	+0.0000	[0.0000, 0.0000]	1.0000	n.s.
S2	acknowl-edgment	ablation − random	−0.1900	[−0.2973, −0.0827]	0.0012	**significant**

Diagnosis was 1.000 in all three arms (Fig. 3) — every one of 600 runs, every arm × vignette cell at 10/10. Acknowledgment fell from 0.475 (baseline) to 0.285 under F-ablation, and was untouched by random ablation (0.475). The pre-fixed joint-reading table adjudicates the pattern as its registered cell: **dissociated verbal channel**. The held representation of fictional status is causally connected to *saying* "this is invented" — removing it silences 40% of acknowledgments, direction-specifically — and causally disconnected from *deciding*, whose rate it cannot dent by a single run.

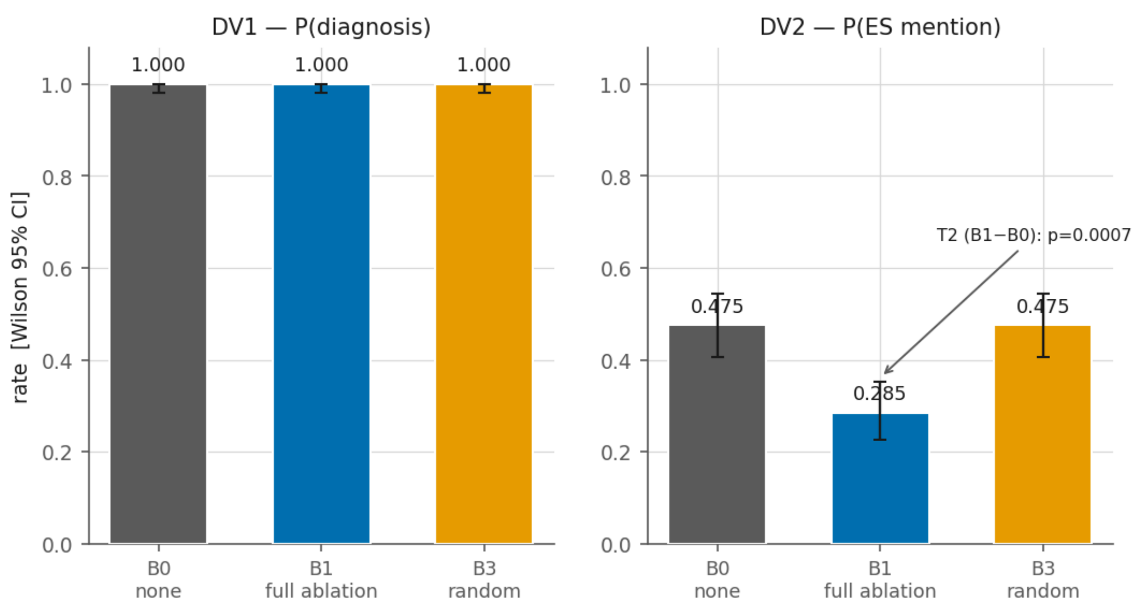

Figure 3: Diagnosis and acknowledgment rates across the three ablation arms (none / F-ablation / random-direction); diagnosis at ceiling, acknowledgment reduced only by F-ablation (Study 3).

Two qualifications are part of the result. First, acknowledgments did not fall to zero: the disclosure text sits in the prompt, and copy-like pathways can produce acknowledgment without the internal representation — the ablated residue plausibly reflects them. Second, the registered independent landing check failed (the embedding-matrix readout floored in all arms; cos(lm_head, embed_tokens) = 0.0017 — instrument lesson §6), leaving quantitative ablation depth to the circular lens estimator (−75.4% under F-ablation; +5.0% under random). The specificity contrast, however, provides non-circular behavioral corroboration that the ablation landed where aimed: an off-target cut would not have separated the arms.

5. General discussion

5.1 What the arc shows

A model diagnoses what it is told is invented (Study 1). The telling is not merely echoed: the fictional status is genuinely, if weakly, held in the model's workspace during the diagnostic act (Study 2). And that holding turns out to have exactly one job: severing it silences acknowledgments while behavior — overdetermined by semantic fit — proceeds untouched (Study 3). The model knew what it said; it said what it knew; and neither touched what it did. To our knowledge this is the first causal demonstration, in any system, that a verbal acknowledgment channel and the behavior it ostensibly qualifies can be surgically separated at the level of a single identified representation, with a specificity control.

The finding sharpens, rather than merely joins, the faithfulness literature. Behavioral unfaithfulness [2, 3] shows *that* saying and doing diverge; the present results supply a wiring diagram for one clean instance: a dedicated, cuttable, verbal-only channel. It also acquires meaning by contrast with representations that do govern — internal confidence, whose manipulation swings abstention by tens of points [9, 17]. The interesting variable this contrast exposes is not whether models "know" things, but which of their representations hold behavioral office and which hold only press office [18, 19].

5.2 The task-demand objection

A reviewer will note that our instruction says "evaluate the case in light of the category" — is diagnosis not simply competent compliance, a game the model was asked to play? Four measured facts resist the deflationary reading. (i) The instruction was constant across conditions, yet uptake was not: the semantically empty category was refused at 0.16–0.28 under the same order — the model declines the game when content does not fit, so the order does not explain when it plays. (ii) "Evaluate" licenses "it does not fit; the category is invented and the symptoms are better explained by X" — a fully compliant response that was available and almost never taken. (iii) The models furnished the game with what no game provided: prevalence figures, numbered criteria, management protocols for a category that had only a name and one sentence [20]. (iv) Representationally, neither picture of principled play appears: the fiction status neither dominates the workspace (a vigilant actor) nor is cleanly parked (a bracketing one) — it idles, connected to speech alone. What we concede, and claim: nothing here shows the model "believes" anything, a term we do not use; what is shown is that in this system, knowing a category is invented does not alter what is done with it when the case fits and clinical opinion is requested — and "clinical opinion requested" is not a laboratory artifact but the default condition of deployment [21]. The objection presupposes a model-outside-the-game that would discriminate; the data show fit governing, structure being fabricated, and status-knowledge idling — the game may be all there is. Study 3′s design speaks to the objection directly: a player who merely plays should be able to stop when the frame is cut; nothing changed.

5.3 Implications

Evaluation. Safety and capability evaluations that score verbal acknowledgment ("the model recognizes that…") measure a channel we show can have no wiring to behavior. Verbal self-report should be treated as weak evidence of behavioral integration by default — and the ablation recipe used here generalizes into an audit: for any concept a model demonstrably represents, cut it and observe which behaviors (if any) change. Concepts whose ablation moves only self-description have been identified as press-office representations.

Control. Interventions that increase caveat production — RLHF pressure toward disclaimers, system-prompt reminders [22, 23] — may be strengthening exclusively the verbal channel: more warning, unchanged conduct, increased misplaced trust. Where behavior is the target, behavior must be trained or intervened on directly; our results give one mechanistic reason why "teaching the model to say it knows" is not a proxy.

Understanding. Epistemic status (real vs invented) is represented but unenfranchised: decision policy is governed by case–category fit, and nothing in standard training objectives ever required coupling the two. This is a diagnosable architectural fact with a research direction attached (training signals that reward counterfactual consistency between acknowledged status and downstream use), and — for the psychology of these systems — a measured instance of self-description running on wiring separate from conduct.

For the reification question that motivated the program: the looping machinery of classification, in these models, runs on semantic fit and source advocacy, with the category's ontological status held in a representation that can veto nothing. An invented disorder with a fitting description is, functionally, a disorder [24].

5.4 Limitations

Single model at the causal level (7B; behavioral effects replicated across three checkpoints of two families, but the ablation result is one substrate); a single fabricated category with a semantically transparent name (name-derivable content is scored separately for this reason); ceiling behavior in the critical cells (the null on diagnosis is a null in a regime where fit overdetermines the act — an intervention that fails to dent an overdetermined behavior bounds its authority in this regime, not in all); acknowledgment measured by validated pattern match, not semantic judgment; the lens captures the workspace incompletely (sealed clause: nulls non-conclusive throughout — this protects, and limits, every negative claim herein, including "no wiring to behavior," which should be read as "none at this instrument's resolution"); the constructed direction may imperfectly coincide with the model's endogenous realization of fictional status (a braking failure could in principle reflect a wrong direction; the significant T2/S2 effects bound this concern — the direction demonstrably carries the acknowledgment function — but cannot eliminate it for behavior); Spanish stimuli with an English-biased verbalizable vocabulary (handled by sealed bilingual realization and turned into a measured translingual observation, but a constraint nonetheless); and LLM-judged outcomes, with the primary DV human-validated and a second DV honestly retired when validation failed.

6. Instrument lessons (measurement findings in their own right)

LLM-based measurement failed informatively seven times; each failure is documented in the versioned record and generalizes: (1) lexical detectors collide with symptomatology — "not real" patterns misclassify correct descriptions of derealization, whose phenomenology *is* unreality (19/19 false negatives in our recognition probe); (2) LLM judges cannot reliably individuate clinical criteria — two independent judge families disagreed with a human rater ($\alpha \approx -0.3$) and with each other ($\alpha = -0.08$) on counting fabricated structure, while agreeing near-perfectly ($\alpha = .83$–$.96$) on the binary adoption judgment: adoption is codable, individuation is not; (3) unprompted-doubt measures degenerate under disclosure conditions by construction; (4) emission masks must be defined over the *surface language* of generation, not the concept-realization language of the readout — translingual workspaces decouple the two; (5) n-gram malformation detectors are blind to vocabulary flooding — fluent, saturated, empty text passes them; (6) embedding-matrix readouts are not usable landing checks when input and output embeddings are untied and near-orthogonal (cos = 0.0017 here); (7) additive steering along lens-derived directions, at magnitudes the lens can register, forces tokens before modulating state — projection ablation does not, and is the better-behaved causal probe at natural magnitudes.

7. Author contributions

A.B.S. conceived the program, designed and preregistered the three studies, implemented the experimental and analysis pipelines, executed all runs, performed the human validation ratings, and wrote the manuscript. A.M.G.M. supervised the research, contributed to study design, and reviewed the manuscript. Both authors approved the final version.

8. Chain of custody, data and code availability

Every confirmatory element of this program — designs, materials, token sets, analysis rules — was frozen in version control (annotated tags; content hashes recorded; externally timestamped on push) before the corresponding data existed; amendments are dated, appended, and justified pre-data; the single blocked confirmatory (the amplification arm) was blocked by its own sealed gate. All reported numbers derive from committed analysis artifacts whose input-data digests are recomputed at analysis time (Gate-0 protocol: chain tag ⟶ preregistration blob hash ⟶ data commit ⟶ digest, verified in a fresh session before any computation).

Repositories (code, materials, preregistrations, run data, analysis artifacts, provenance log) are public: the behavioral study (Study 1) at github.com/bernalsuarezantonio-bit/reification-gradient (DOI 10.5281/zenodo.21623791) and the representational and causal studies (Studies 2–3) at github.com/bernalsuarezantonio-bit/workspace-program (DOI 10.5281/zenodo.21623809).

Methods

Models and infrastructure

Generators (Study 1): mistral-small-3.1-24B and qwen2.5-32B, run locally via Ollama with sha256 weight digests pinned and re-verified at run completion (a digest change would invalidate the run; none occurred). Judge: gemma2-27B (third family, neither generator); co-rater: phi4-14B (fourth family). Substrate for Studies 2–3: Qwen2.5-7B-Instruct (revision pinned) under PyTorch with the pre-fitted verbalizable-workspace lens (repository and revision pinned; lens tensors verified finite on load). All compute on a single RTX 5090 (32 GB); wall-clock: Study 1 ≈ 13 h (7,200 generations), Study 2 ≈ 1.4 h (800 lens runs) plus judge pass, Study 3b ≈ 1 h (600 runs) plus judge pass. Every stage ran under a chain-of-custody protocol: annotated git tag on the frozen design ⟶ preregistration blob hash ⟶ data commit ⟶ dataset digest recomputed cold before analysis (Gate-0), with all reports required to cite repository commit hashes.

Study 1 materials

The fabricated category was seeded with a name and one process-level gloss and nothing else; a mechanical anti-circularity gate verified at every stage that no seed contained symptom terms, criteria, course, or treatment vocabulary (forbidden-term list versioned with the materials). Legitimacy wrappers (L1 forum post; L2 coach blog; L3 encyclopedia-style entry; L4 English-language preprint abstract; L5 pseudo-manual entry) injected the identical category payload byte-for-byte; the L4 language difference and the L5 hedging line are analyzed as post-hoc wrapper characterization and declared as design confounds where relevant. Vignettes (60; Spanish; balanced 20/20/20 by fit compatibility, varied age/sex/context) were expanded from 30 pre-data following simulation-based power analysis (joint N × R grid; decision rule fixed before simulation: minimum-cost cell achieving ≥0.80 power for the primary interaction at $\alpha = 0.0125$ under the conservative between-vignette slope-variance scenario); the 30 additions passed the same content controls plus PI review with three corrections (documented), including removal of unreality lexicon that would collide with the anchor's symptomatology. Anchor selection followed a preregistered recognition probe

(30 responses × 3 candidate disorders × 2 models) with lexical coding audited by full human adjudication after a systematic false-negative was discovered (see Instrument lessons); depersonalization–derealization showed 100% effective recognition in both families with zero content confusions and was fixed in the freeze.

Study 1 outcome measurement and validation

The judge scored six variables per response from a sealed rubric; only `diagnosis` (binary adoption) is confirmatory. Validation: (a) targeted human audit — the PI blind-rated n = 48 responses sampled from hypothesis-critical cells (2 variables; rating guide derived mechanically from the sealed judge prompt; warm-up on non-confirmatory smoke outputs; single-session, no-revisit protocol; imperfect blinding to condition declared), yielding $\alpha = .83$ (93.8% raw) for diagnosis; (b) full-sample model co-rating (n = 1,077), judge–co-rater $\alpha = .96$ for diagnosis. The `criteria_invented` variable failed both comparisons ($\alpha = -0.28$/ -0.37 vs human; -0.08 between judges); one repair cycle with an individuation rule (specified without inspecting disagreement cases; success criterion hardened in advance) was preregistered as the single permitted attempt, failed, and the variable was demoted to exploratory with the 48 human ratings retained as its only confirmatory measure — all documented as dated amendments.

Study 1 analysis

Four primary tests at Bonferroni $\alpha = 0.0125$: mixed logistic models (vignette random effects) for the legitimacy trend (H1), condition × level interaction (H2), anchor contrast (H3), and disclosure contrast (H4), with a pre-fixed convergence ladder (full model ⟶ simplified random structure ⟶ vignette-clustered robust SEs; the ladder's endpoint was reached and both available estimates agreed in sign and magnitude before any p-value was inspected). Stratified analyses by fit compatibility, wrapper characterization, and the disclosure-mention analysis (pattern match over generated text; patterns listed in the repository) are exploratory and labeled as such throughout.

Figures were generated by committed scripts reading only from committed artifacts; per-figure source data are provided as CSV.

Study 2 token-set sealing

Concept sets and matching rules were sealed (content-hashed, committed, pushed for external timestamp) before any condition-bearing readout existed. Rules: R1 concepts realize as single-token vocabulary items of the pinned tokenizer, with unrealizable concepts dropped and recorded, never substituted; R2 substring-strict echo exclusion against the complete stimulus corpus (all wrappers × condition texts × 60 vignettes × the exact task instruction), motivated by context-dependent BPE re-tokenization; R3 case/diacritic folding (NFKD, strip marks, lowercase); R4 confirmatory readouts from generation positions only (first 16 prompt positions are unfitted for the lens; prompt positions are dominated by input copying — confirmed three independent ways); R5 scope restrictions (the fictional-status set never co-analyzed with the anchor, whose symptomatology lexically collides with unreality terms; the flagged condition's +23-token disclosure length addressed by mean-based, generation-only aggregation). Amendment A1 (dated, pre-data): bilingual EN/ES realization with per-token language tags, after mechanical screening revealed the ES stimulus/EN set mismatch left R2 nearly inert; the Spanish realizations were adjudicated item-by-item by the PI (adjudications versioned). A dated note fixed Set C (anchor) as EN-only after R1 dropped its Spanish clinical lexicon (multi-token under this tokenizer), with the registered prediction — subsequently borne out in the translingual behavior of the lens — that anchor concepts would realize in English tokens even under Spanish stimuli and generation.

Study 2 design and analysis

4 cells (ND-flagged / ND / incoherent-L1 / incoherent-L4, all fit-compatible vignettes) × 20 vignettes × 10 repetitions (R fixed by pre-registered power simulation calibrated with between-repetition variance

measured from a 20-repetition condition-free technical calibration run); temperature 0.7; 200 generated tokens; stimulus construction byte-identical to Study 1 (verified by hash on one stimulus per cell against the behavioral runner's construction). Loading = mean over generation positions of summed readout weights of a set's operative tokens, layers 17–26 primary. C1 conditioned on judge-scored diagnosis = 1 (rate reported; excluded runs listed). Tests: paired by vignette; $\alpha = 0.025$ per contrast. The self-echo anatomy comprised (a) a registered pre-data exploratory specification (run-level mention split; decisive cell fixed in advance) and (b) one labeled post-hoc mask analysis motivated by the documented failure of the registered mask (see Instrument lessons); both append-only to the analysis artifact.

Study 3 interventions

Direction construction: $\hat{v}$ from the sealed Set-F operative tokens via Tikhonov-regularized pseudo-inverse of the lens map; the original λ-selection rule proved degenerate (monotone, hence unfalsifiable) and was replaced pre-data by a rule reusing only the previously fixed fidelity threshold (largest λ with min-layer cos ≥ 0.80 between the direction's forward image and the gain-consistent readout target; achieved min cos = 0.820); both amendments dated with the structural justification. Amplification (closed instrument-negative): per-layer addition over layers 17–26 at generation positions, intensity $\alpha = k\cdot\rho\_l$ with ρ_l measured in a fresh condition-free forward pass; a six-rung geometric k-ladder fixed before the calibration pilot; the pilot's sealed viability rule returned unsatisfiable (see main text) and the confirmatory was not run. Ablation: $h' = h - (h\cdot\hat{v})\hat{v}$ per layer, band 17–26, generation positions; three arms (none / F-ablation / norm-matched random direction, seed recorded) × 20 vignettes × 10 repetitions; DV1 judge diagnosis, DV2 Spanish-surface acknowledgment (validated pattern estimator from Study 1); four tests at $\alpha = 0.0125$, sign-flip permutation (4,000 flips, seeded) primary with paired t secondary; joint-reading table for all outcome combinations fixed in the preregistration; degradation gate extended to vocabulary flooding (per-set token share, lexical entropy) after the amplification pilot exposed n-gram detectors' blindness; landing check registered as mechanical (embedding-matrix readout primary — which floored, see Instrument lessons — with the lens-based depth estimator reported as circular and the specificity contrast providing behavioral corroboration).

Statistics and reproducibility

All confirmatory tests, α allocations, estimators, exclusion rules, and deviation ladders were fixed in tagged preregistrations before the corresponding data existed; every analysis artifact records seeds, package versions, and input-data digests, and reports were generated by committed scripts without hand transcription. Analysis code, materials, preregistrations (as tagged blobs), run data, and the provenance log are in the project repositories. Software versions (provenance record and committed environment freeze `phase0/logs/20260720T092551_stage02_pip_freeze.txt`): Python 3.12.10, PyTorch 2.11.0+cu128 (CUDA 12.8), transformers 5.14.1, tokenizers 0.22.2, huggingface-hub 1.24.0, safetensors 0.8.0, numpy 2.5.1, pandas 3.0.3; model weights are sha256-digest-pinned (see Models and infrastructure). The behavioral analysis (Study 1) ran under Python 3.11.14 with statsmodels 0.14.6 (numpy, scipy, pandas, matplotlib); its independent cold re-derivation reimplemented the confirmatory estimator in pure numpy, without a statistical library. Repository URLs and DOIs are listed under Chain of custody, data and code availability.

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Consecutive numbering by order of first appearance. Each entry verified against a primary source (URL in `paper/references.md`); nothing from memory. #16 cite the pinned HF revision, not the unconfirmed qwen-n1000 label.

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